The periodic table of elements 5a

A brief history

In 1829 scientists knew about 40 elements. German scientist Johann Dobereiner noticed that lithium, sodium and potassium had very similar properties. Calcium, strontium and barium were also similar to each other. Chlorine, bromine and iodine formed another triad. In 1864, English scientist John Newlands replaced the triads with a law of octaves, which was based on the atomic weights of 60 known elements as well as their properties. Dmitri Mendeleev, from Russia, refined the periodic table in 1869, leaving gaps for some elements which had not been discovered. Mendeleev predicted the properties of some of those missing elements quite accurately. However, there were some inconsistencies: certain elements were too heavy for their position in the periodic table. Henry Mosely (England, 1914) resolved these problems, when he based the order of the elements on their atomic number rather than their weight. Today we know 103 elements and they are all organised into the periodic table of elements.

Some properties of the first 20 elements

Name	Symbol	Atomic weight	Order by weight	Atomic number	Atomic radius (pm)	Melting point (°C)
Aluminium	Al	27		13	143	660
Argon	Ar	40		18	94	– 189
Beryllium	Be	9		4	112	1280
Boron	B	11		5	88	2030
Calcium	Ca	40		20	197	838
Carbon	C	12		6	77	3730
Chlorine	Cl	35		17	99	–101
Fluorine	F	19		9	64	–220
Helium	He	4		2	50	–270
Hydrogen	H	1		1	37	–259
Lithium	Li	7		3	152	1650
Magnesium	Mg	24		12	160	650
Neon	Ne	20		10	60	–249
Nitrogen	N	14		7	70	–210
Oxygen	O	16		8	66	–219
Phosphorus	P	31		15	110	44
Potassium	K	39		19	231	64
Silicon	Si	28		14	118	1410
Sodium	Na	23		11	186	98
Sulfur	S	32		16	102	113

Note: pm = picometre = 10^{-12}m = 1/ 1 000 000 000 000 metres

1 Order the elements by weight and write their order numbers into the appropriate column above.

2 State at which point (by element name) this order is different to the atomic number order.

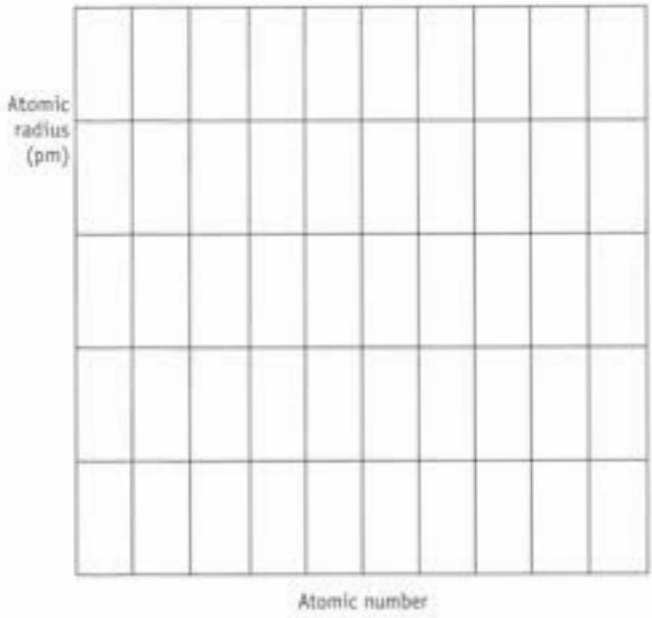
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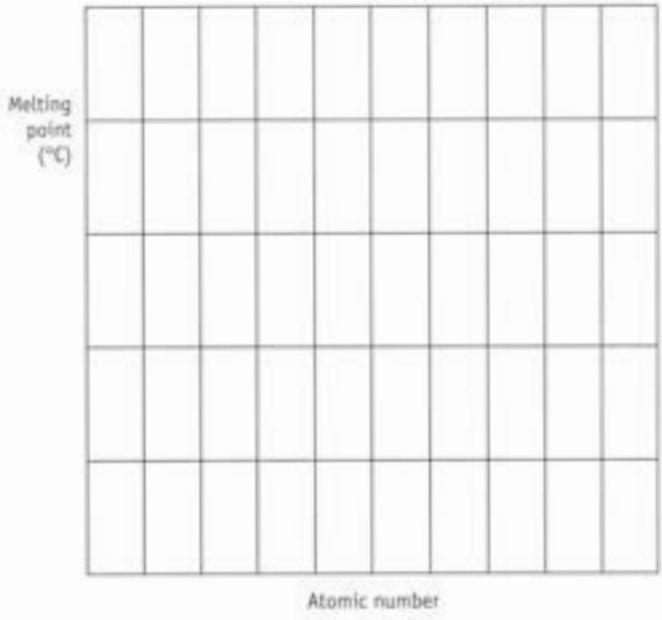
3 Graph the atomic radius on the vertical axis against the atomic number on the horizontal axis on the first grid on the next page.

4 Graph the melting point against the atomic number on the second grid.

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The periodic table of elements 5b





The periodic table for the first 20 elements

5 Complete the periodic table for the first 20 elements below by writing the symbols into the spaces. Let the atoms with a large atomic radius begin the horizontal periods. Let the elements with a low melting point finish the periods. Let the other elements fit into the periodic table in order of their size and melting point. The first element, hydrogen, is often written above the periodic table, because it has properties from the first and the seventh horizontal group.

						1	2
3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18
19	20						

Trends in the periodic table

Form a diagonal line across the periodic table by shading in the cells of boron and silicon. To the left of this line are the metals, to the right of the line are the non-metals. The shaded elements are semi-metals.

6 Complete the table below by summarising some general properties of metals and non-metals.

Aspect	Metals	Non-metals
Conduct electricity?		
Conduct heat?		
Are solid?		
Are malleable?		

7 Complete the table below by answering 'yes' or 'no' about changes in size, weight and properties of atoms.

Property	Within periods (going across)	Within groups (going down)
Atomic radius increases		
Atomic weight increases		
Atoms have similar properties		

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